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Press Information

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Civic 04 - 3 Door Type R 2004

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The Civic 3 door Type-R

The Civic Type-R continues in the footsteps of the Integra and Accord Type-R derivatives, delivering exceptional dynamic ability, innovative engineering and an exhilarating driving experience in a unique road car package.

Striking, purposeful looks are matched by a high output, 2.0 litre engine, which features Honda's advanced DOHC i-VTEC technology. Applied to the Civic Type-R, output rises to a prodigious 200 PS (147 kW) at 7,400 rpm, with peak torque of 196 Nm delivered at 5,900 rpm.

100 PS/litre equates to a top speed of 235 km/h, while a 30 per cent lighter flywheel and

improved engine response for the MY 2004 Type-R have brought the 0-100 km/h acceleration time down from 6.8 to an even more potent 6.6 seconds; yet combined cycle fuel consumption remains at just 8.9 l/100 km.

i-VTEC is the generic name of Honda's outstanding petrol engine family. The name is derived from 'intelligent' combustion control technologies that match outstanding fuel economy, cleaner emissions and reduced weight with high output and greatly improved torque characteristics in all speed ranges.

The design cleverly combines the highly renowned VTEC system - which varies the timing and amount of lift of the valves - with VTC or Variable Timing Control. VTC is able to advance and retard inlet valve opening by altering the phasing of the inlet camshaft to best match the engine load at any given moment. The two systems work in concert under the close control of the engine management system delivering improved cylinder charging and combustion efficiency, reduced intake resistance, and improved exhaust gas recirculation among the benefits. i-VTEC technology offers tremendous flexibility since it is able to fully maximise engine potential over its complete range of operation.

A particularly flat torque curve is testimony to its effectiveness: by 3,000 rpm the engine is already delivering in excess of 180 Nm.

The 2.0 litre DOHC i-VTEC engine in detail
Salient features are:
* DOHC VTEC
* VTC or Variable Timing Control
* compact intake system
* rear exhaust system with low heat mass
* camshafts operated by a particularly quiet chain drive
* a serpentine belt providing auxiliary drive similar to that used in the Honda S2000, keeping the engine compact
* ladder frame main bearing stiffener located between the block and sump, increasing rigidity

VTEC
The principle of the VTEC system is to optimise the amount of air-fuel charge entering, and the amount of exhaust gas leaving, the cylinders over the complete range of engine speed to provide good top-end output together with low and mid-range flexibility. Typically, at high engine speeds the valves remain open for a longer duration to give the gases sufficient time to overcome their inertia. At low and mid range engine speeds, where valves opening for too long would allow intake charge to leak back out and exhaust gases to leak back into the cylinder, the valves remain open for a shorter duration.

The switch to the high lift, long duration operation at high engine speeds is achieved by locking the rocker arms operating each pair of inlet and each pair of exhaust valves to a third high lift rocker arm (one on the intake side, one on the exhaust side) by means of a hydraulically-activated pin.

Variable Timing Control (VTC)

In the Civic Type-R's 2.0 litre engine, the VTEC mechanism is complemented by Variable Timing Control that works by taking into consideration engine load.

Based on input from a position sensor located at the rear end of the inlet camshaft, together with a whole range of other data, the engine control unit varies the inlet camshaft position relative to that of the exhaust camshaft by means of a hydraulically driven, compact vane-type pump located on the front end of the inlet camshaft. In this way it can advance and retard the opening of the inlet valves.

During the periods of high engine load which occur during acceleration, VTC is set at a relatively small degree of valve overlap which provides the optimum output, the valve opening angle fully utilising the inertia effect of the intake air. In addition, as engine speed builds, the VTEC mechanism switches from the low speed cam to the high-speed cam (i.e. optimal torque to optimal power), but with the same degree of overlap.

In contrast, at high engine speeds in situations where the engine is not under heavy load, for example during motorway cruising, there is much greater valve overlap and this is advantageous in reducing pumping losses, maximising the exhaust gas recirculation effect for reduced NOx levels and providing the best balance between fuel consumption and output.

Finally, at idle and low engine speeds during light load conditions, inlet valve opening is retarded for minimal overlap, generating strong swirl and therefore good mixing. EGR is reduced and this stabilises combustion.

Compact dimensions, thermally efficient

On the 2.0 litre engine, the stainless steel exhaust manifold is mounted on the back of the engine. This means the distance to the catalytic converter is reduced, so less heat is lost. Also the 4-2 exhaust manifold flows into an exhaust pipe structure which features an 'e-pattern' cross section rather than a dual pipe construction, which is even more heat efficient.

6-speed manual transmission

Complementing the Type-R's DOHC i-VTEC engine is a 6-speed close ratio manual gearbox. High performance synchronisers and the high efficiency changing mechanism allow extremely quick and precise gear changes.

The transmission is matched to a high performance clutch and features triple cone synchronisers on both first and second gears. Shorter in length than many 5-speed transmissions, it offers an exhilarating shifting feel as well as superior packaging. Carefully spaced ratios maintain engine revs well within the power band during acceleration.

Chassis

The inherently fine-handling Civic 3 door with its highly rigid body and much praised suspension design provides an ideal platform for a high performance derivative. Nevertheless, Honda has introduced additional stiffening to endow the Type-R model's handling with greater tautness and precision.

High rigidity

At the rear, a strut fitted between the wheelarches, together with a reinforced wheelarch gusset increases vertical dynamic rigidity by 23 per cent. Frontal horizontal rigidity benefits to the tune of 17 per cent thanks to an additional strut located at the base of the front bulkhead and between the two front side members.

This is matched by firmer dampers and springs, uprated anti-roll bars front and rear (that at the front is stiffer, that at the rear is increased in diameter compared to standard 3 door models) and 17 x 7JJ alloy wheels shod with 205/45 R17 tyres. Backed up by a programme of extensive testing at Germany's demanding Nurburgring, the package ensures that the Civic Type-R possesses excellent chassis dynamics, with a high degree of linearity in its handling behaviour.

Handling and stability benefit from a 15 mm lower ride height and uprated anti-roll bars: a higher gauge steel at the front, but using the same 25.4 mm diameter, and a diameter increased from 13 to 18 mm at the rear.

Larger front disc brakes than the standard Civic 3 door, with a firmer pedal action, provide stopping ability to match.

Exterior

To ensure a high degree of aerodynamic efficiency, the designers used a computer-generated 'virtual wind tunnel' to predict wind flow around the car and every aspect of the body was studied to reduce drag and minimise wind noise. This included developing the general shape of the body, refining the front spoiler, adding a rear suspension cover, and even refining the shape of the door mirrors.

Styling revisions, while adding to the Type-R's overt, sporty appeal, are more than just cosmetic. Each of the additional body panels - chin spoiler, side sill garnish, rear under spoiler, larger roof spoiler - have been carefully shaped and tested to provide improved aerodynamic performance.

Nevertheless, in combination with a 15 mm lower ride height and the squat, wide track design of the Civic 3 door, the looks leave no doubt as to the high performance intent of the Type-R model. The effect is set off by a mesh-type front grille complete with 'Type-R' script, black-plated headlamp sub-reflectors and twin chrome tail pipes. Even under the bonnet there is no mistaking the performance potential: the red crackle-finish cam cover is picked out with the legend 'DOHC i-VTEC'. For 2004, brake callipers now incorporate the Type-R logo for added exclusivity. Door mirror-mounted indicators and three light projector style headlamps complete the picture.

Interior

The aggressive sporting image is reflected inside the car. Large, racing car-influenced seats are firmer and provide even more support than those of the standard 3 door model, with the addition of bolsters at shoulder height and those along the seat base sides increased in depth by 10 mm.

The Type-R's performance credentials are further signalled by white instrument faces and an aluminium-effect gear knob; additional flourishes are provided by an embroidered Type-R logo at the base of the headrest and red stitching on both the seats and the steering wheel. Door and seat inserts, as well as headrest apertures, are now finished in red.